How long should a block be?

One World Extremes

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Block maximum method

Idea promoted for engineering applications by Emil Gumbel (cf. Gumbel 1958).

- divide a (time) series of length b into n_b equal-length disjoint blocks of b successive observations,
- fit the generalized extreme value (GEV) distribution to the resulting sample of n_b block maxima,
- use the fitted model for risk assessment.

Motivating example

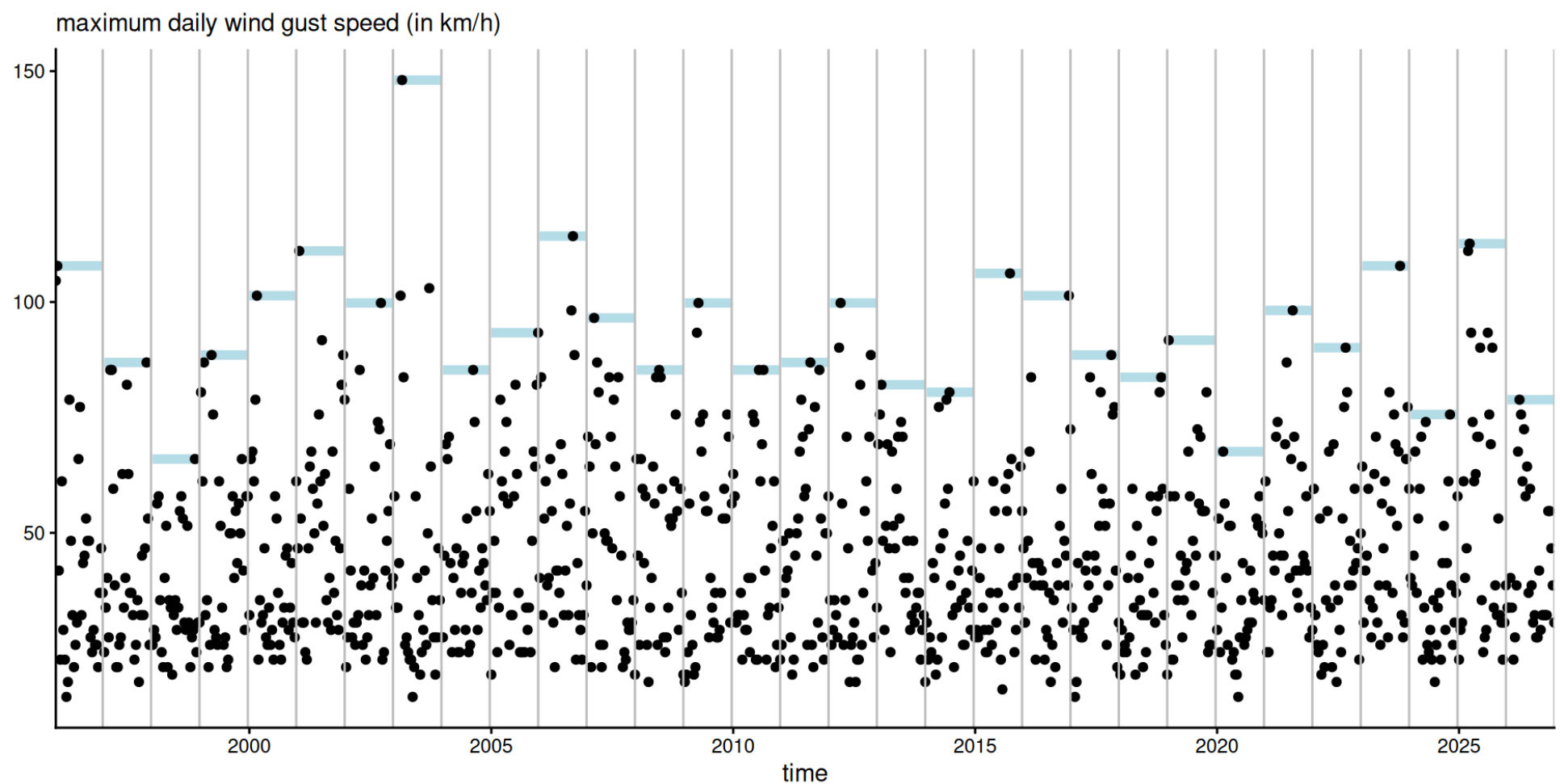


Figure 1: Daily maximum wind gust speed in Cheeseboro, California, for the months of January (plus Feb. 1st) 1996–2026, for a total of 32×31 measurements.

Extremal types theorem

The justification for using the GEV comes from Fisher and Tippett (1928), Gnedenko (1943), von Mises (1954).

If G is a distribution function and there exist location and scaling sequences d_m and $c_m > 0$ such that the maximum of m observations has a non-degenerate limiting distribution

$$\lim_{m \to \infty} G^m(c_m x + d_m) = F(x)$$

then $F(x)$ must be a GEV distribution function.

Generalized extreme value distribution

The GEV distribution function is

$$F(x;\mu,\sigma,\xi) = \begin{cases} \exp\left[-\left\{1+\xi\frac{x-\mu}{\sigma}\right\}_+^{-1/\xi}\right], & \xi \neq 0, \\ \exp\left[-\exp\left\{-\frac{x-\mu}{\sigma}\right\}\right], & \xi = 0. \end{cases}$$

with location $\mu \in \mathbb{R}$, scale $\sigma > 0$ and shape $\xi \in \mathbb{R}$.

The support of $\mathsf{GEV}(\mu,\sigma,\xi)$ is $\{x : \sigma + \xi(x-\mu) > 0\}$.

Max-stability

The GEV is characterized by *max-stability*:

if $X_j \overset{\text{iid}}{\sim}$ GEV(μ, σ, ξ), then for any m,

$$Y = \max\{X_1, \ldots, X_m\} \sim \text{GEV}(\mu_m, \sigma_m, \xi),$$

with

$$\begin{aligned}
\mu_m &= \mu + \sigma(m^\xi - 1)/\xi, & \sigma_m &= \sigma m^\xi, & &\text{if } \xi \neq 0,\\
\mu_m &= \mu + \sigma \log m, & \sigma_m &= \sigma, & &\text{if } \xi = 0.
\end{aligned}$$

Stationary setting

The extremal types theorem extends to stationary time series (Leadbetter 1974, 1983; Leadbetter et al. 1983).

Under a condition $D(u_n)$ and for extremal index $\theta \in (0, 1]$, we get instead

$$\lim_{m \to \infty} G^{\theta m}(c_m x + d_m) = F(x)$$

or equivalently

$$\lim_{m \to \infty} G^{m}(c_{\theta m} x + d_{\theta m}) = F(x).$$

Research question: How to choose the block size m?

Traditional extreme value bias-variance trade-off:

- Smaller block sizes may lead to poor GEV approximation (potentially biased extrapolation).
- Larger blocks lead to smaller sample sizes (more variability).

Other considerations

- Data availability (frequency of records).
- Block maximum is widely viewed as more robust approach in the applied literature.
- Theoretical consideration of asymptotic efficiency relative to peaks-over-threshold (Bücher and Zhou 2021).
- Natural (minimum) block size (e.g., annual for seasonally-varying data).

Literature review

Surprisingly, very little work in the area.

- Wang et al. (2016) suggest using combinations of goodness-of-fit tests (such as Anderson–Darling or Kolmogorov–Smirnov).
- unpublished work of Dkengne et al. (2020) suggests parameter stability plot built using max-stability.

Diagnostic plots

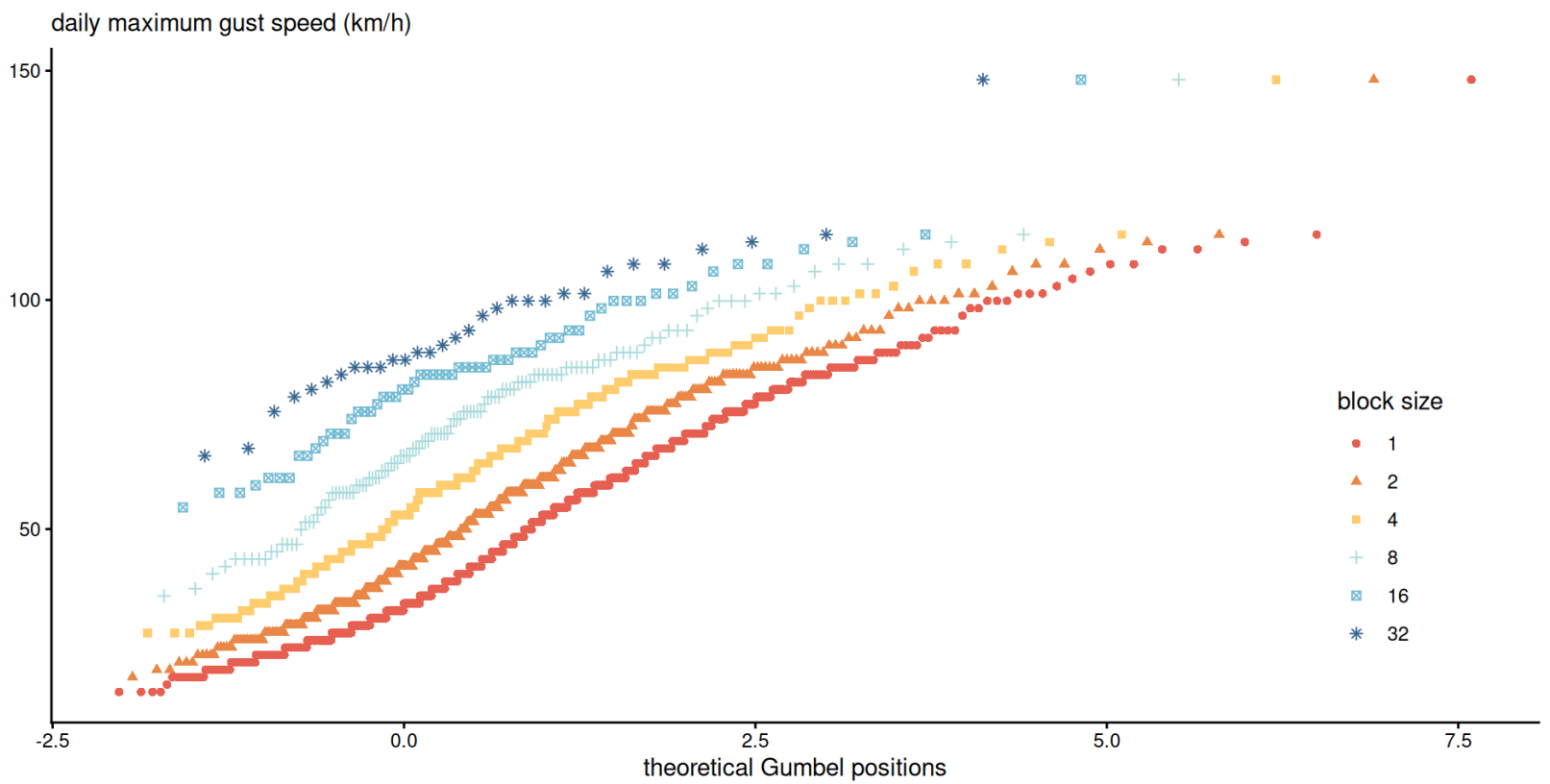


Figure 2: Gumbel plot of m-day maxima against standard Gumbel plotting positions $-\log[-\log\{i/(n+1)\}]$ for the Cheeseboro wind gust data.

Plot *à la* Cox et al. (2002). With Gumbel fit, lines should be parallel, but we see some evidence of distortion for $b = 1$ or $b = 2$ days.

Testing for different block sizes

Consider blocks of size b versus bm for $b, m \in \mathbb{N}$. We gather observations into an $n_b/m \times m$ matrix of ordered inputs.

Table 1: Blocks of size $m = 4$ from daily maxima wind gusts (in km/h).

23	42	105	108
14	23	29	61
18	32	48	79
21	26	31	66
32	43	45	77

Row i contains the order statistics $X_{i,(1)} \leq \cdots \leq X_{i,(m)}$.

Joint distribution of order statistics

Let $X_{(1)} \leq \cdots \leq X_{(m)}$ denote the order statistics.

Let F and f be the CDF and density of $X_j \overset{\text{iid}}{\sim} \text{GEV}(\mu, \sigma, \xi)$.

The joint density of order statistics is the product of

- the conditional $X_{(1)}, \ldots, X_{(m-1)} \mid X_{(m)} = x_{(m)}$,
- the marginal $X_{(m)}$.

Conditional density of lower order statistics

The conditional density of $X_{(1)}, \ldots, X_{(m-1)} \mid X_{(m)}$ is

$$f_{(1,\ldots,m-1)|(m)}(\boldsymbol{x}_{-(m)}) = (m-1)! \prod_{j=1}^{m-1} \frac{f(x_{(j)}; \mu, \sigma, \xi)}{F(x_{(m)}; \mu, \sigma, \xi)}$$

so the $(m-1)$ smallest observations are, up to permutation, $\mathsf{GEV}(\mu, \sigma, \xi)$ truncated above at $x_{(m)}$.

Distribution of block maximum

The block maximum $Y = \max(X_1, \ldots, X_m)$ has CDF $F_{(m)}(x) = F^m(x)$ and density

$$\begin{aligned} f_{(m)}(x) &= mF(x; \mu, \sigma, \xi)^{m-1} f(x; \mu, \sigma, \xi) \\ &= f(x; \mu_m, \sigma_m, \xi), \end{aligned}$$

by max-stability.

Likelihood

We write

$$\mathcal{L}(\boldsymbol{\vartheta}_0, \boldsymbol{\vartheta}_m) = \prod_{i=1}^{n} f_{(1,\ldots,m-1)|(m)}(\boldsymbol{x}_{i,-(m)}; \boldsymbol{\vartheta}_0) f_{(m)}(x_{i,(m)}; \boldsymbol{\vartheta}_m),$$

where

- $\boldsymbol{\vartheta}_0 = (\mu, \sigma, \xi)$ are the parameters for $X_{i,(1)} \leq \cdots \leq X_{i,(m-1)}$.
- $\boldsymbol{\vartheta}_m$ are the parameters of the largest observation $X_{i,(m)}$.

Under the null hypothesis of max-stability,
$\boldsymbol{\vartheta}_m = (\mu_m, \sigma_m, \xi) = g(\boldsymbol{\vartheta}_0)$, say.

Alternative hypotheses

The idea underlying our testing procedure is to let the parameters of the block maximum differ, by

- A_1: setting $\mathcal{H}_a : \boldsymbol{\vartheta}_m = (\mu_{\omega m}, \sigma_{\omega m}, \xi)$ for some scaling $\omega > 0$, and testing $\mathcal{H}_0 : \omega = 1$.
- A_2: allowing both location and scale to vary $\mathcal{H}_a : \boldsymbol{\vartheta}_m = (\mu_m + \nu, \phi\sigma_m, \xi)$ and testing $\mathcal{H}_0 : \nu = 0, \phi = 1$.
- A_3: letting all three parameters vary $\mathcal{H}_a : \boldsymbol{\vartheta}_m = (\mu_m + \nu, \phi\sigma_m, \xi + \zeta)$ and testing $\mathcal{H}_0 : \nu = 0, \phi = 1, \zeta = 0$.

We use likelihood ratio statistics to compare the models $\mathcal{H}_0$ and $\mathcal{H}_a$. The asymptotic distribution of the test is χ^2_k with $k \in \{1, 2, 3\}$.

Power study stup

We consider different scenarios to assess the power of tests:

- sample sizes $n \in \{25, 50, 100\}$,
- number of replicates $m \in \{2, 5, 10\}$,
- three alternatives $\mathcal{H}_a$.

All power curves are based on 2000 replications, with tests conducted at level 5%.

Power study

We consider maxima of

a. **normal**$(0, 1)$ and

b. **Weibull**$(1, 0.8)$.

with distribution G and block of size 30, simulate

- $X_{(1)}, \ldots, X_{(m-1)}$ from G^{30},
- $X_{(m)}$ from $G^{30\delta}$, but left-truncated at $x_{(m-1)}$.

Power study - generalized extreme value data

We use the penultimate approximation to get GEV parameters and simulate from these. Both have limiting shape $\xi = 0$, but convergence is slow.

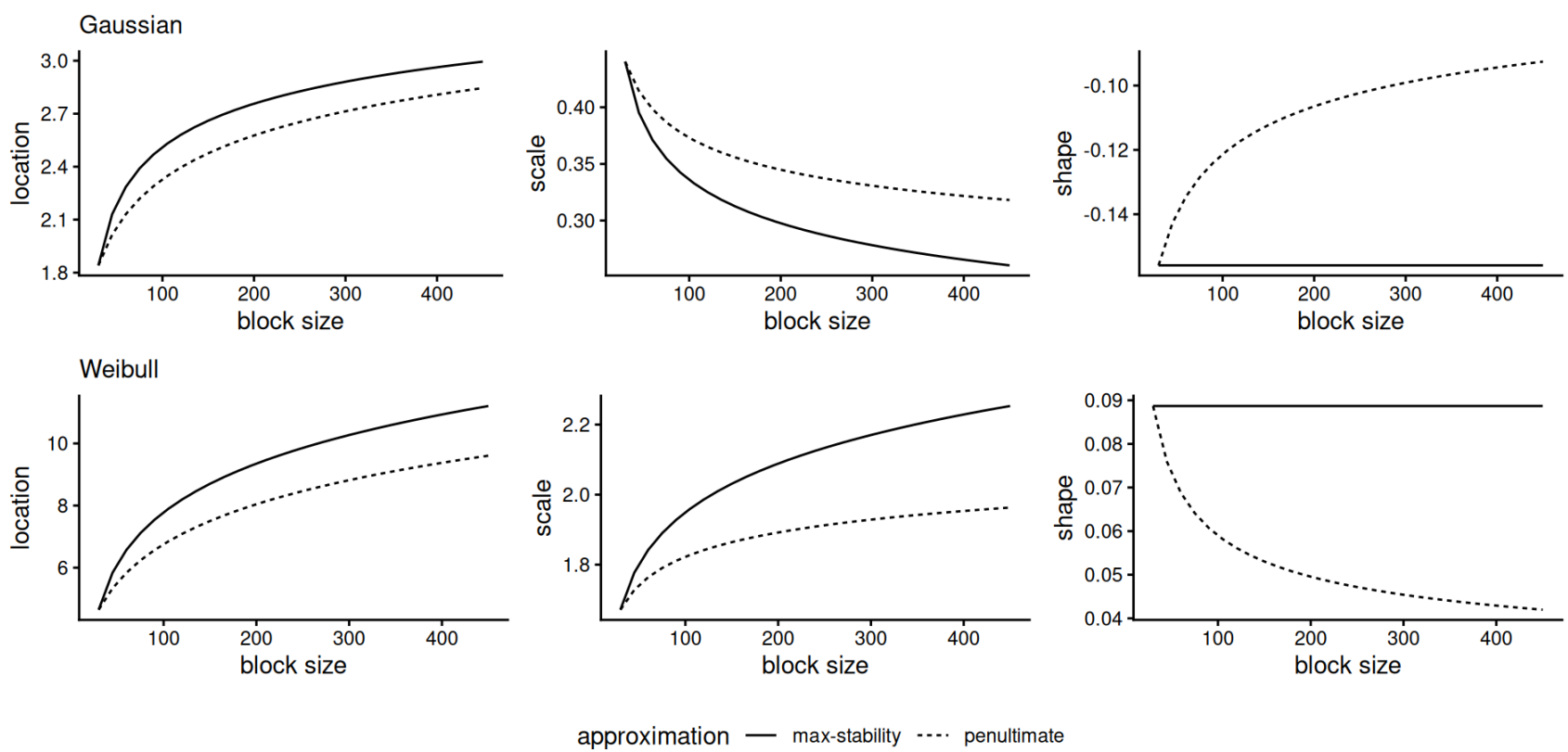
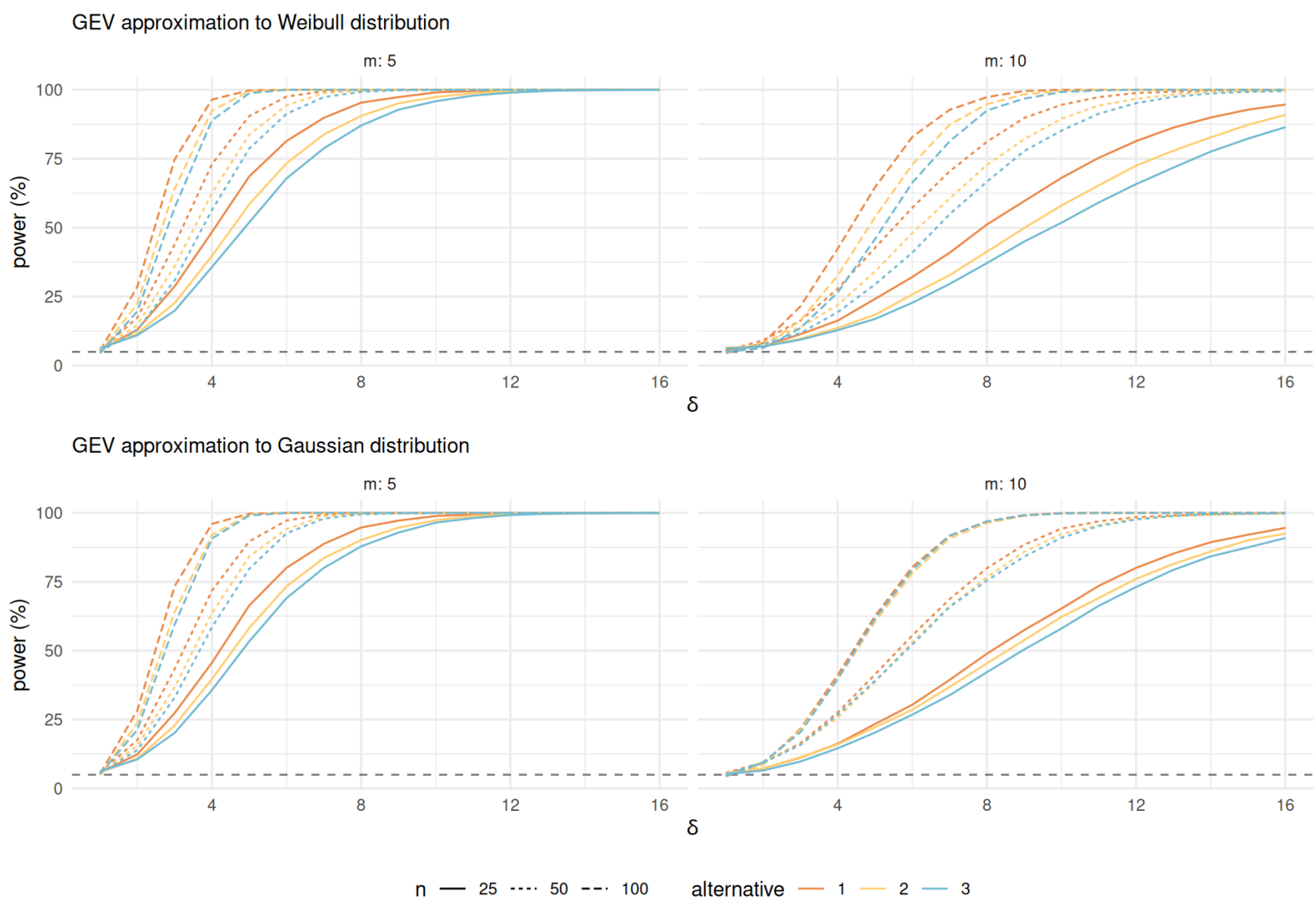
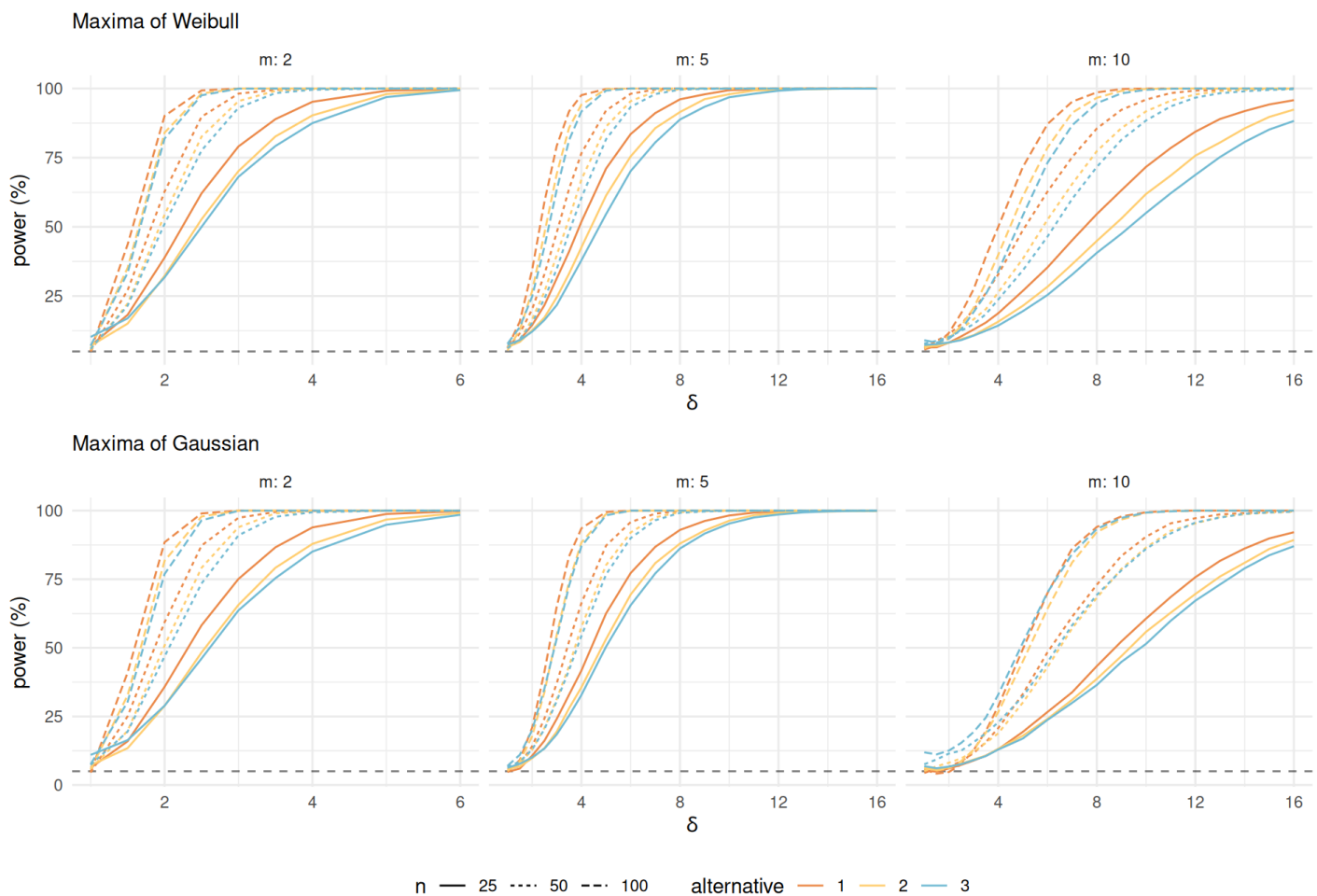


Figure 3: GEV parameters based on max-stable (dashed), and the penultimate approximation (full) for Gaussian (top) and Weibull (bottom).

Power curves - GEV



Power study - maximum domain of attraction



We simulate directly block maxima from G^{30}, where G is the Gaussian or Weibull CDF, but where the largest observation is drawn from $G^{30\delta}$.

Summary of power study

- As expected, power increases with the sample size n.
- Perhaps surprisingly, the test has more power when m is small!
- Alternative 1 is always the more powerful (less degrees of freedom), despite misspecification.
- There is a small size distortion (about 5%) for alternative 3 in small samples.

Stationary sequences with extremal clustering

To investigate serial dependence, we also consider data from first-order max-autoregressive (MAR) processes parametrized in terms of the extremal index θ.

Consider the stochastic process obtained through the recursion (Tavares 1977)

$$
\begin{aligned}
X_i &= \max\{X_{i-1}, Z_i\} + \log(1-\theta), \quad i \geq 1 \\
Z_i &\sim \mathsf{GEV}\{\log(\theta) - \log(1-\theta), 1, 0\}
\end{aligned}
$$

The margins are standard Gumbel, $X_i \sim \mathsf{GEV}(0, 1, 0)$, while the distribution of $Y = \max\{X_1, \ldots, X_m\}$ is Gumbel with location $\log[(m-1)\theta + 1]$.

Location parameters of stationary sequences

Max-stability holds asymptotically, but with $m\theta$ rather than m.

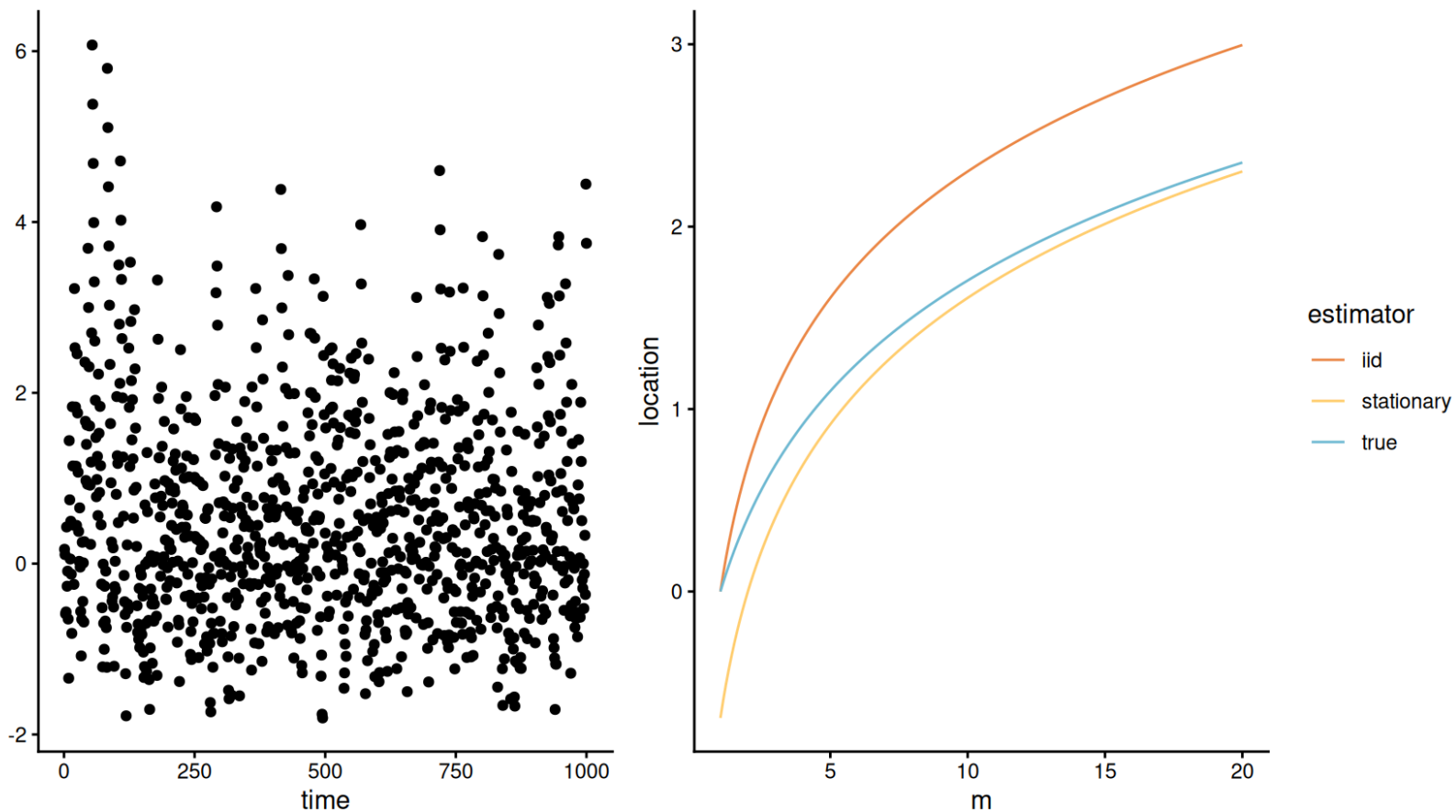


Figure 4: MAR with $\xi = 0$ and $\theta = 0.5$: time series of simulated values (left) and location parameters of the Gumbel maxima based on extrapolating to blocks of size m, taking $\log(m)$ (iid) and $\log(m\theta)$ (stationary) as location parameters.

Max-autoregressive process with positive shape

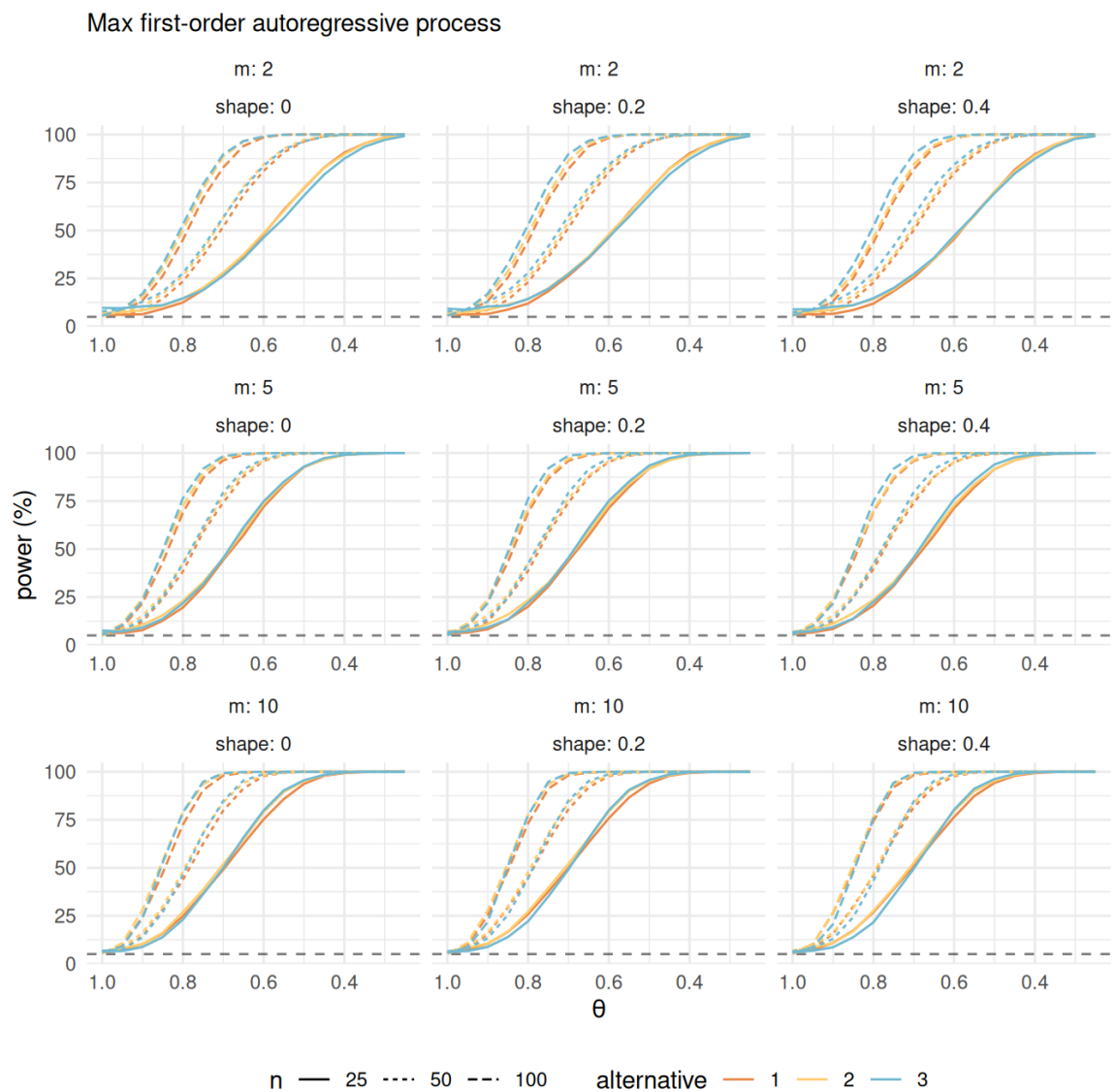
For $\xi > 0$, the first-order max-autoregressive model is

$$X_i = \max\{(1-\theta)^{\xi} X_{i-1}, Z_i\}, \qquad i \geq 1$$
$$Z_i \sim \text{GEV}(\theta^{\xi}, \xi\theta^{\xi}, \xi)$$

The marginal stationary distribution is Fréchet with shape ξ^{-1}, i.e., a GEV$(1, \xi, \xi)$.

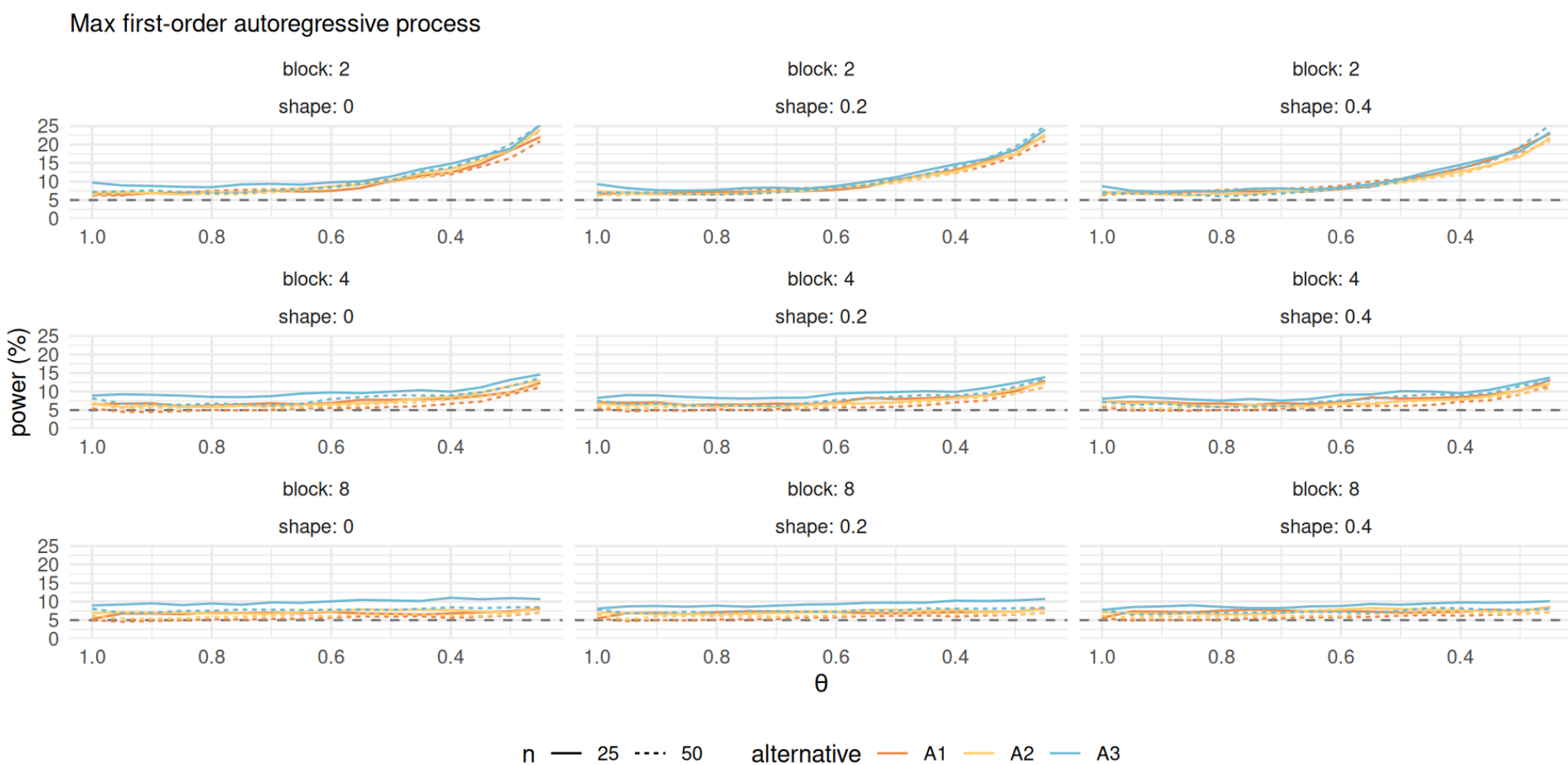
The distribution of $Y = \max\{X_1, \ldots, X_m\}$ has CDF $\exp[-\{\theta(m-1)+1\}x^{-1/\xi}]$; see Example 10.3 of Beirlant et al. (2004).

Power curves - max autoregressive



Power study - doubling the block size with MAR

As the empirical block size gets larger, the GEV approximation should become effective and directly incorporate the impact of θ.



Comments on power curves

- More discrepancies are expected for smaller values of θ, as the average cluster size is $1/\theta$, the reciprocal of the extremal index.
- As the block size increases, the effect of θ is captured through parameter estimates and we fail to reject for larger θ, as expected.
- Alternative 3 suffers again from size distortion, especially in small samples (either when m or n is small).

Diagnostic plots

We propose quantile-quantile plots to assess the goodness-of-fit of the model.

Among quantities of interest

- all maxima,
- the block maximum $X_{(m)}$,
- the spacing $X_{(m)} - X_{(m-1)}$.

We use a parametric bootstrap scheme for uncertainty quantification.

Pivots

Under the null hypothesis, the $F(X_j; \boldsymbol{\vartheta})$ are uniform and standard results on order statistics of uniform samples gives for the spacing

$$S(\boldsymbol{\vartheta}) = F(X_{(m)}; \boldsymbol{\vartheta}) - F(X_{(m-1)}; \boldsymbol{\vartheta}) \sim \mathsf{beta}(1, m)$$

and for the largest observation

$$F(X_{(m)}; \boldsymbol{\vartheta}) \sim \mathsf{beta}(m, 1).$$

Alternative pivots can be obtained, e.g., via Rényi's representation.

Parametric bootstrap

A conventional parametric bootstrap to test properties of the statistic $S(\boldsymbol{\vartheta})$ would

- simulate uniform draws $U^*_{(1)} \leq \cdots \leq U^*_{(m)}$.
- map them to GEV via the quantile function $X^*_j = F^{-1}(U^*_i; \widehat{\boldsymbol{\vartheta}})$ using the MLE $\widehat{\boldsymbol{\vartheta}}$, letting $X^*_j \sim \text{GEV}(\widehat{\mu}, \widehat{\sigma}, \hat{\xi})$.
- obtain maximum likelihood estimates $\widehat{\boldsymbol{\vartheta}}^*$ from $\boldsymbol{X}^*$.
- use the distribution of the back-transformed values of $X^*_{(m)}$ and $X^*_{(m-1)}$, e.g., $S(\boldsymbol{X}^*; \widehat{\boldsymbol{\vartheta}}^*)$
- apply the CDF of the beta distribution to get uniform draws.

Distortion due to plug-in

The null distribution of $S(\widehat{\boldsymbol{\vartheta}})$ depends on the unknown true parameter value $\boldsymbol{\vartheta}$, and although this is replaced by an estimator that is consistent under the null hypothesis, some distortion remains.

The resulting error can be reduced by nested bootstrapping (Davison and Hinkley 1997, sec. 3.9), but this may be too computationally complex for routine use.

Marginal likelihood

Since interest lies in extrapolation, we can also consider estimation of GEV parameters based on the marginal distribution of $X_{(1)}, \ldots, X_{(m-1)}$, where

$$\begin{aligned}\mathcal{L}_{\mathrm{marg}}(\boldsymbol{\theta}) &= \prod_{i=1}^{n} f_{(1),\ldots,(m-1)}(\boldsymbol{x}_i; \boldsymbol{\theta}) \\ &\propto \prod_{i=1}^{n} \{1 - F(x_{i,(m-1)})\} \prod_{j=1}^{m-1} f(x_{i,(j)}; \boldsymbol{\theta})\end{aligned}$$

which amounts to right-censoring the largest observation.

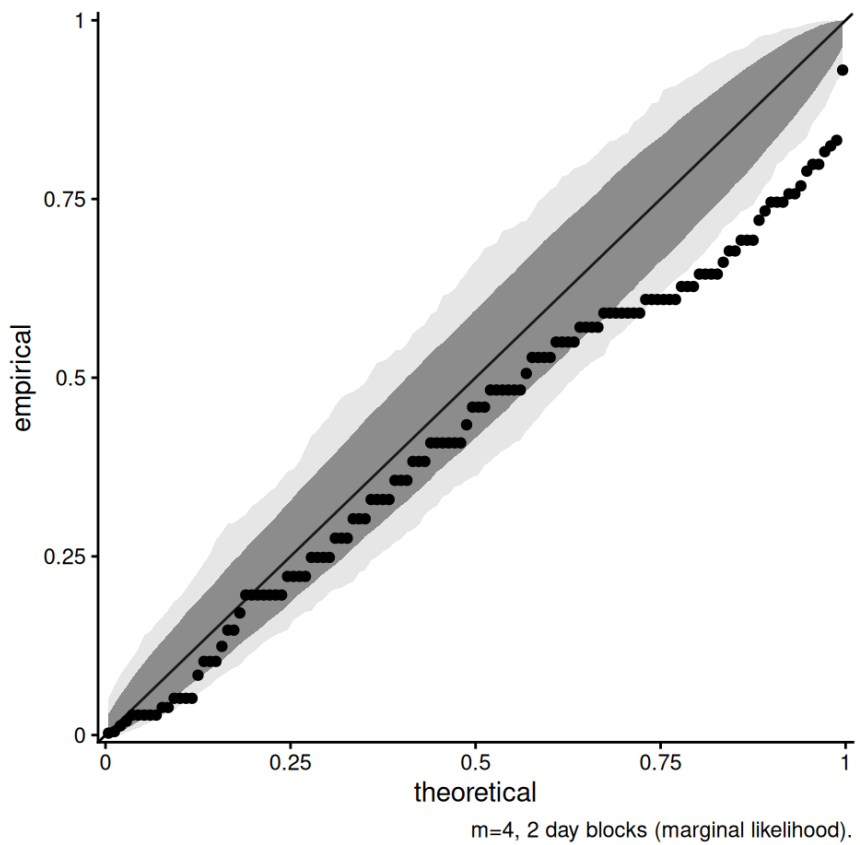
(Dis)advantages of the marginal likelihood

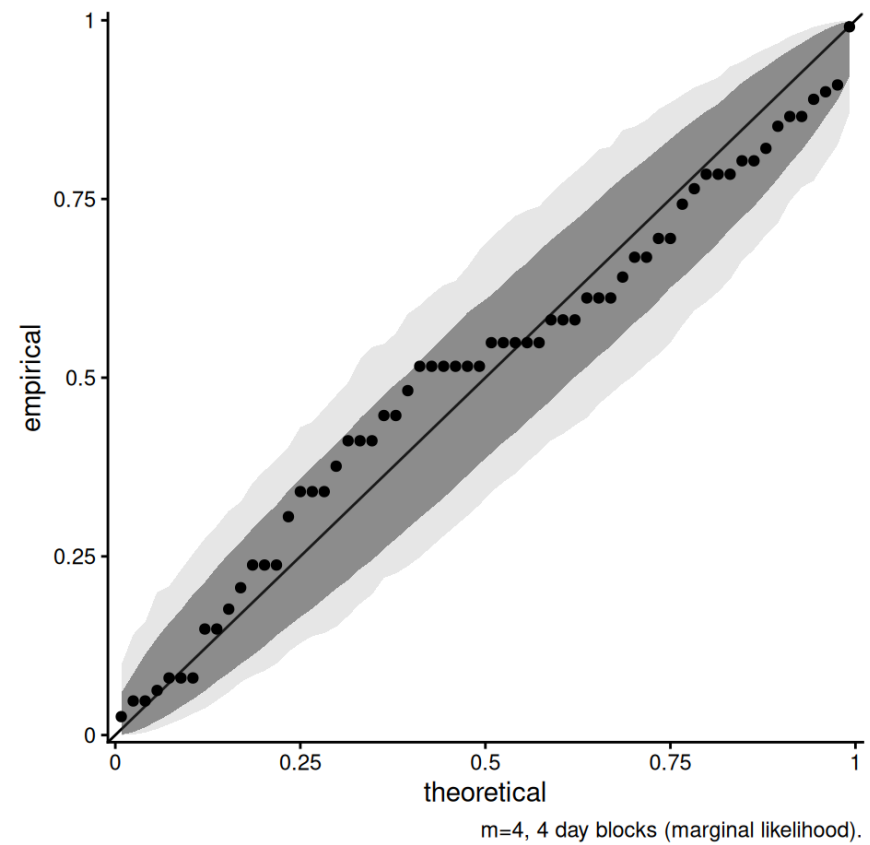
P-values for test for wind gusts at Cheeseboro

Table 2: P-values (rounded to three digits) for the test of max-stability.

	$m = 2$			$m = 4$		
block	A_1	A_2	A_3	A_1	A_2	A_3
$b = 1$	0	0	0	0	0	0
$b = 2$	0.005	0.02	0	0.081	0.21	0
$b = 4$	0.279	0.47	0.16	0.252	0.47	0.17
$b = 8$	0.176	0.38	0.47	0.743	0.76	0.82

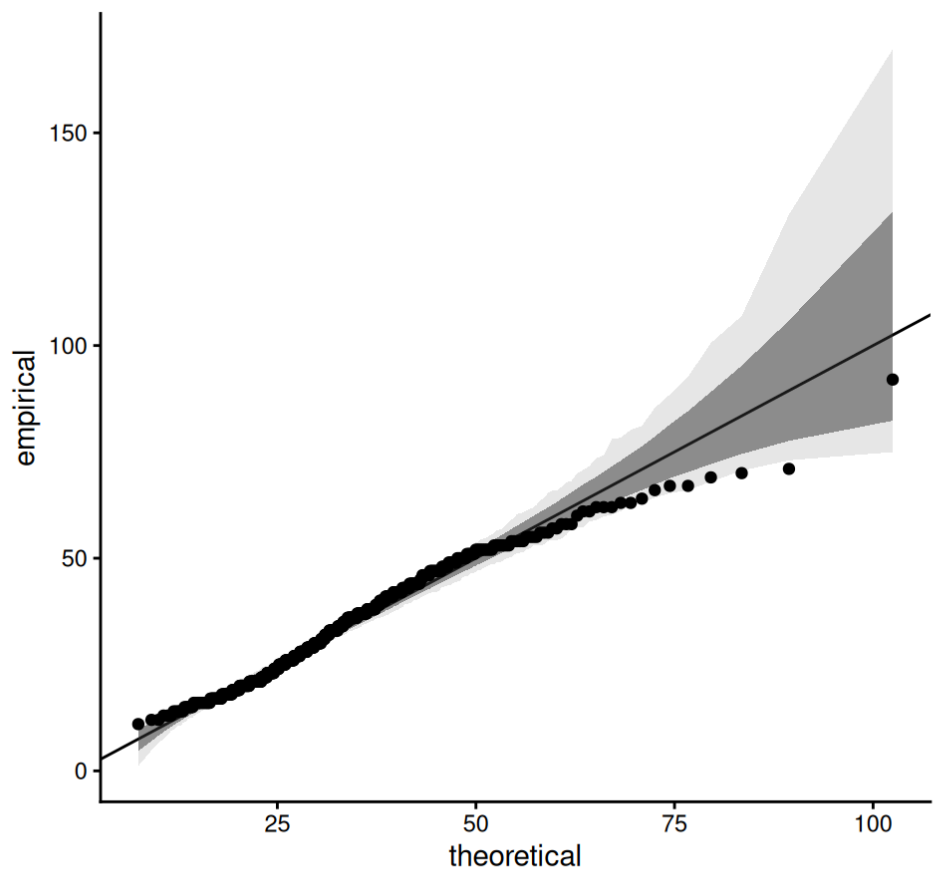
Probability-probability plots of maxima



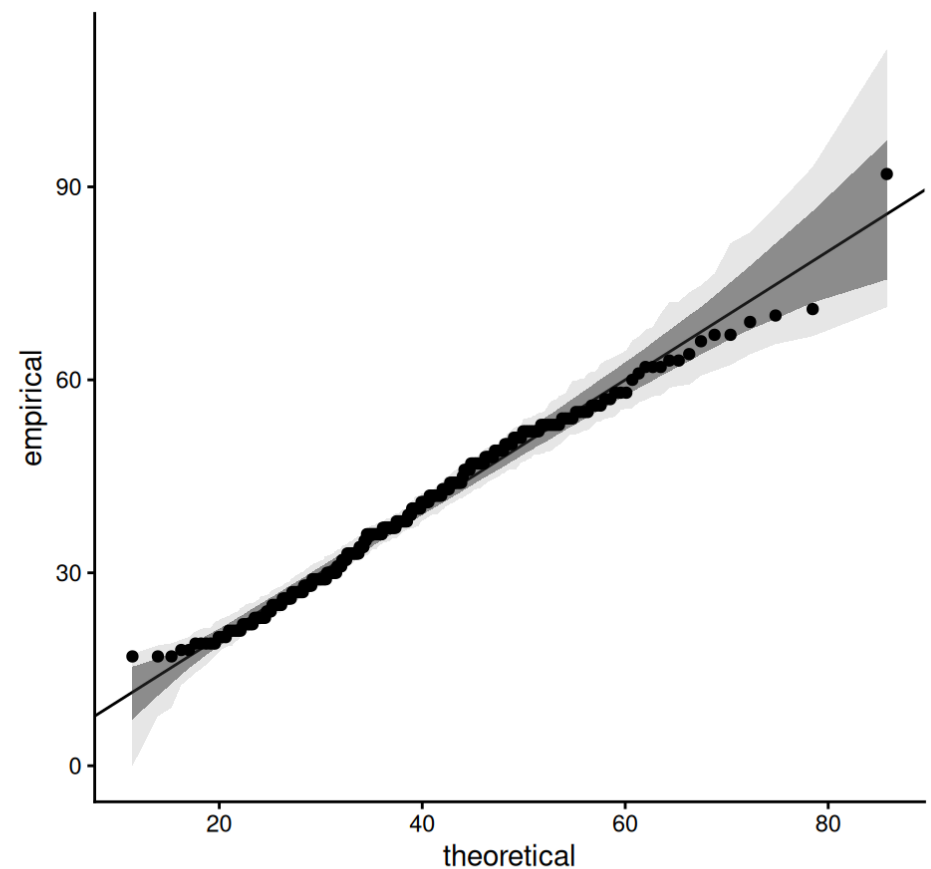


We obtain pointwise and simultaneous confidence intervals using the envelope method (section 4.2.4 of Davison and Hinkley 1997).

Quantile-quantile plot of all maxima



2 day blocks



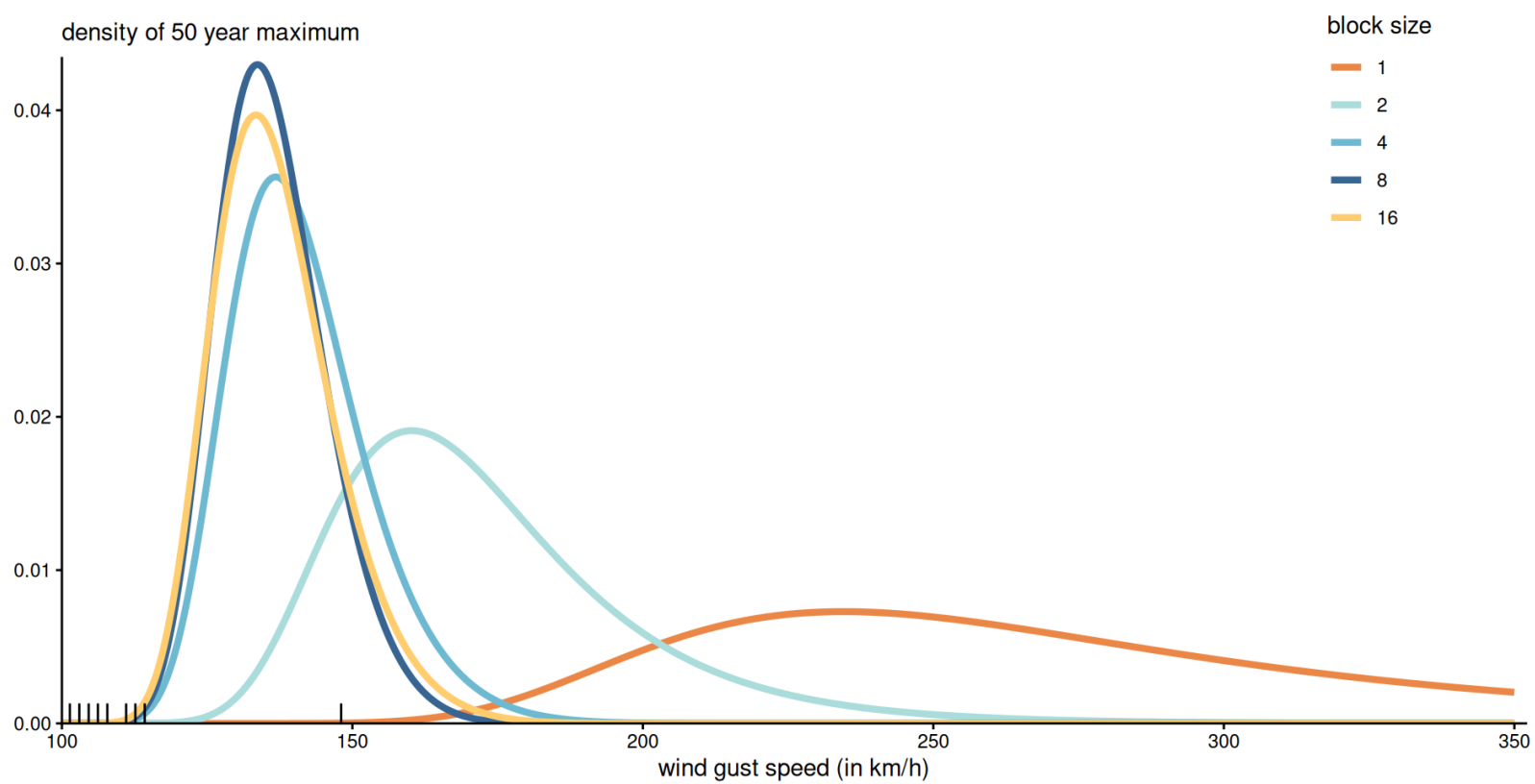
4 day blocks.

Work in progress to account for rounding.

Risk extrapolation

The block size can have a sizeable impact on risk estimation (sample maximum is 148km/h).

Shape of $\hat{\xi} = 0.18$ for $b = 1$, versus $\xi = -0.09$ for $b = 4$.



Density of the 50-year maximum for the Cheeseboro wind gust data, based on different block size and maximum likelihood estimates.

Inference for extremes

Use GEV model fitted to blocks of size 4 days to calculate risk measures for the maximum 50 years wind gust, by using the max-stability relationship with $m = 8 \times 50$.

- The estimated median of half-century maximum daily wind gust is 139.6 km/h
- 95% profile-based confidence intervals: (125.9km/h, 168.5km/h).
- The serial correlation is negligible.

Summary

We propose likelihood ratio tests for testing max-stability for n_b blocks of size b

- Build blocks of $n_b/m \times m$ ordered data.
- Check fit to largest (max of m) and use max-stability to detect departures.
- Simulation studies show the test is powerful even in small samples.
- The test with a single parameter seems more powerful than competitors, more for $m = 2$.

Diagnostic plots based on bootstrap to account for goodness-of-fit.

- Different values give different indications about the fit (all observations, only the max of blocks of size bm, spacings, etc.) give visual indications of goodness-of-fit.

Limitations

- We consider only bm versus b:
  - does not allow for comparison of blocks of size 2 vs 3, say.
  - some data discarded if n is not divisible by bm (but could use ideas from Simpson and Northrop (2026)).
  - in practice it is natural to consider aggregation of periods, e.g., monthly to yearly.
- The scheme leads to multiple testing if multiple values of m and block sizes are considered:
  - smaller values of m should lead to higher power, as shown through simulations,
  - but the choice of m should be dictated more by practical considerations.
- Our approach is restricted to independent or stationary time series.

Extensions and future work

- Accounting for rounding (work in progress) and more guidelines for the choice of graphical diagnostic.
- Combining model with estimates of the extremal index θ.
- Alternative estimators for the GEV built from sliding blocks (e.g., Bücher and Zanger 2023).

Theory for these is more complicated, owing to lack of genuine likelihood.

- Could consider Wald-based tests with sandwich-based or bootstrap covariance estimators.

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